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B3

EXPERIMENT NO. 2

Configuration and Comparison of Network Hardware Components Using Cisco Packet Tracer

EXPERIMENT NO. 02

Configuration and Comparison of Network Hardware Components Using Cisco Packet Tracer

AIM:

To study and configure various network hardware components and analyze their functionality.

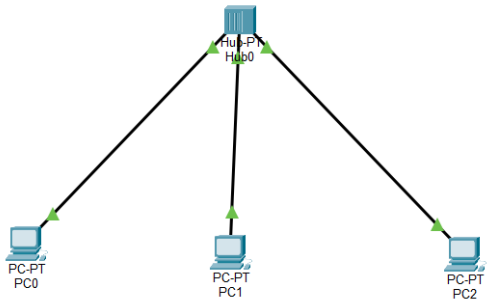
Objectives:

- Configure routers, switches, hubs, and end devices.
- Understand the role of each network hardware component.
- Analyze data flow through network devices.
- Compare device performance and usage scenarios.
- Verify connectivity across network components.

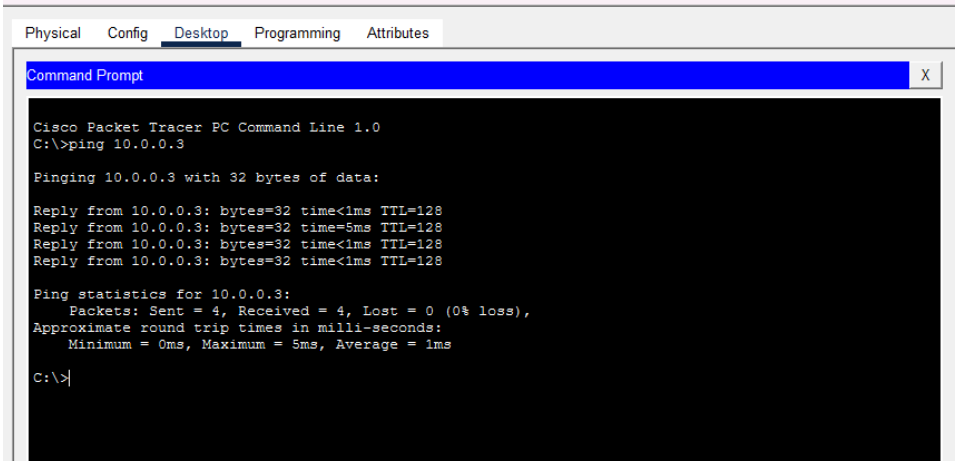
EQUIPMENT/SOFTWARE: Cisco Packet Tracer 7.3.0

TOPOLOGY TO STUDY HUB OPERATION:

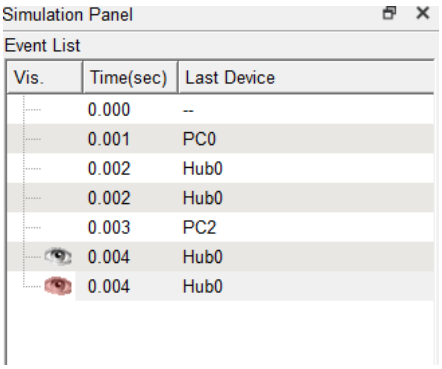
SCENARIO:



PING RESPONSE:

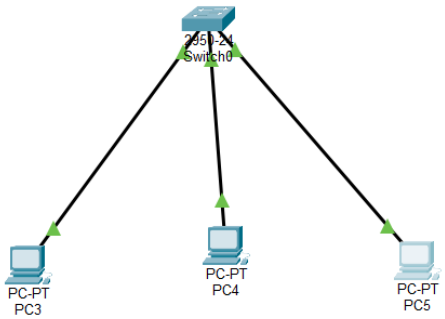


PACKET FLOW SEQUENCE:

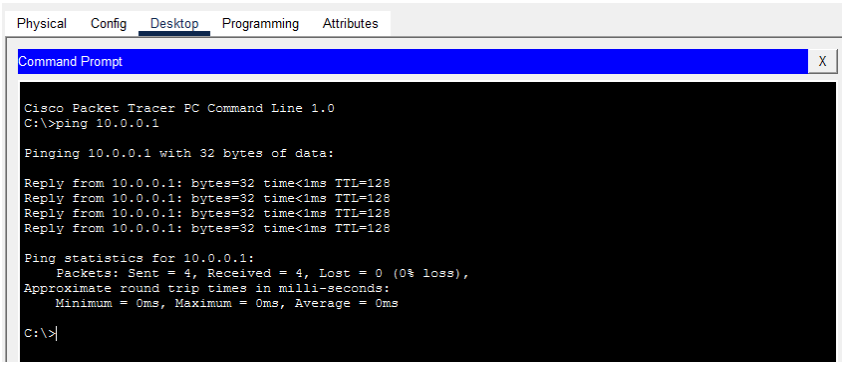


• TOPOLOGY TO STUDY SWITCH OPERATION:

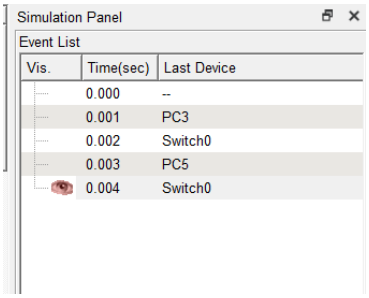
SCENARIO:



PING RESPONSE:

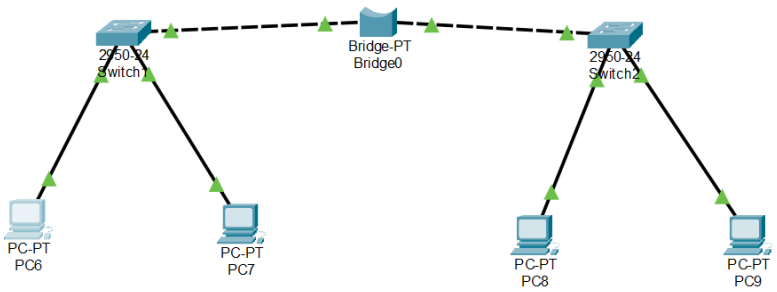


PACKET FLOW SEQUENCE:



TOPOLOGY TO STUDY BRIDGE OPERATION:

SCENARIO:



PING RESPONSE: PC1 TO PC3

```
C:\>ping 10.0.0.4

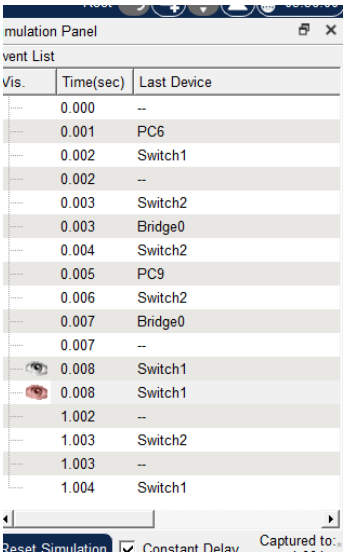
Pinging 10.0.0.4 with 32 bytes of data:

Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time=1ms TTL=128
Reply from 10.0.0.4: bytes=32 time=5ms TTL=128

Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 1ms

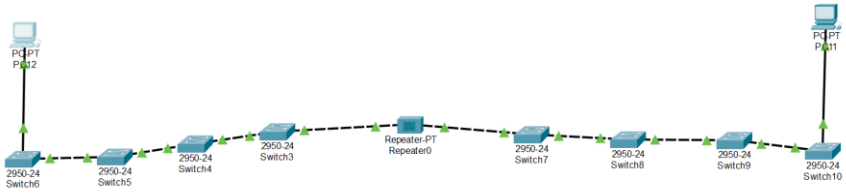
C:\>
```

PACKET FLOW SEQUENCE:

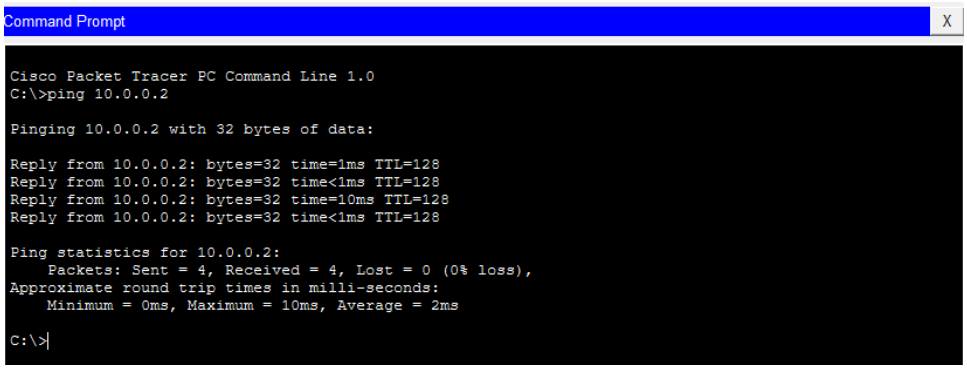


TOPOLOGY TO STUDY THE USE OF REPEATER:

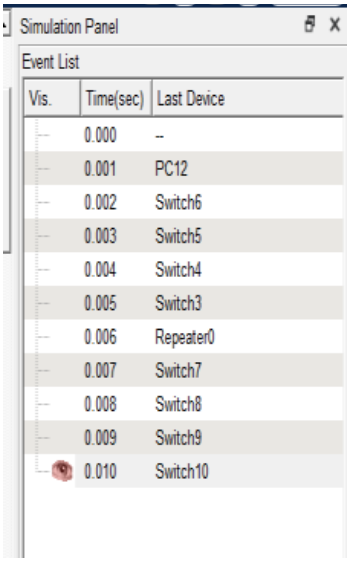
SCENARIO:



PING RESPONSE:

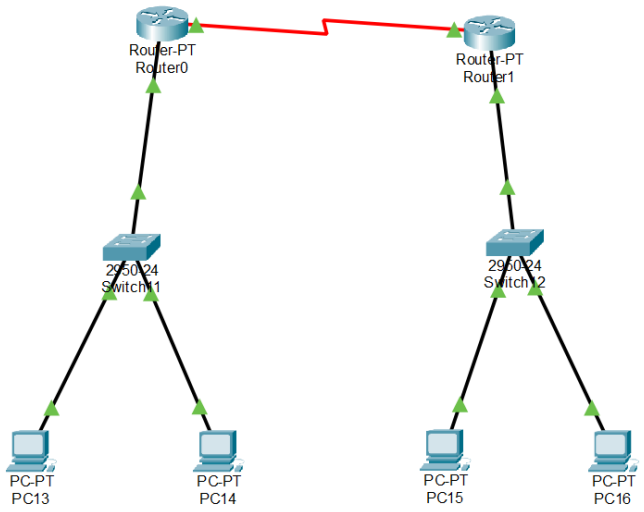


PACKET FLOW SEQUENCE:

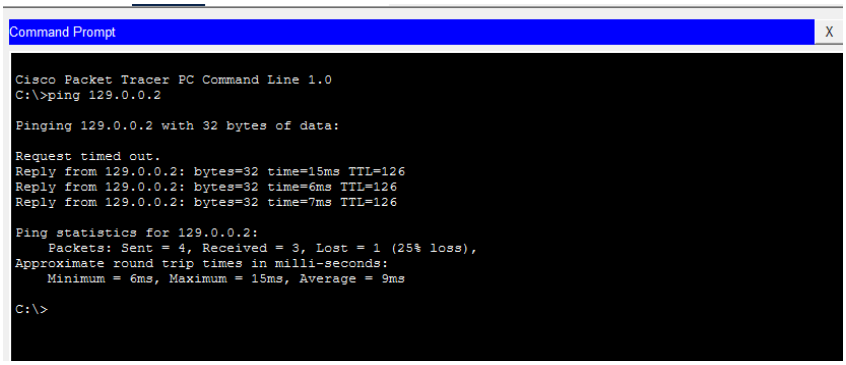


• TOPOLOGY TO STUDY ROUTER OPERATION:

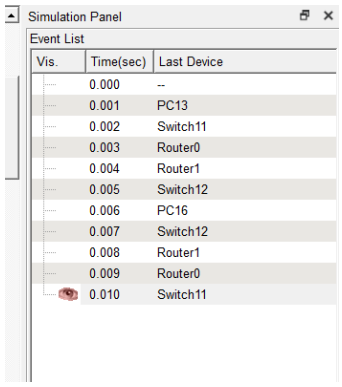
SCENARIO:



PING RESPONSE:



PACKET FLOW SEQUENCE:



COMPARISON BETWEEN ALL THE NETWORK DEVICES:

HARDWARE DEVICES → PARAMETERS	**HUB**	**SWITCH**	**ROUTER**	**BRIDGE**	**REPEATER**
Layer in OSI model	Physical Layer (L1)	Data Link Layer (L2)	Network Layer (L3)	Data Link Layer (L2)	Physical Layer (L1)
Functions	Regenerates and broadcasts incoming signals.	Connects devices within a LAN; forwards frames via MAC.	Connects distinct networks; routes packets via IP.	Connects two LAN segments based on MAC addresses	Amplifies/regenerates electrical signals to extend range.
No. of ports available	Multi-port (Typically 4 to 24 ports)	Multi-port (High density, 24 to 48+ ports)	Low-port density (Typically 2 to 4 routing interfaces)	Low-port density (2 ports)	Low-port density (2 ports)
Filtering capability	No	Yes (Filters frames by MAC address)	Yes (Filters packets by IP/ACLs)	Yes (Filters frames by MAC address)	No
Memory	No	Yes (Content Addressable Memory/CAM Table)	Yes (Routing Table, NVRAM, RAM)	Yes (Bridge/MAC Table)	No

COMPARISON BETWEEN HUB AND SWITCH:

HARDWARE DEVICES → PARAMETERS	**HUB**	**SWITCH**
Layer in OSI model	Physical Layer (L1)	Data Link Layer (L2)
Transmission type	Broadcast (Sends data to all ports)	Unicast / Multicast / Broadcast (Smart delivery)
Table	None	MAC Address Table (CAM Table)
Function	Acts as a simple multiport repeater.	Inspects frame headers and filters traffic dynamically
No. of ports available	4, 8, 12, or 16 ports	8, 16, 24, or 48 ports
Collision Domain	Single collision domain for the whole device.	Per-port collision domain (Each port is isolated).
Transmission Mode	Half-Duplex (Cannot send and receive simultaneously)	Full-Duplex (Can send and receive simultaneously)
Cost	Inexpensive / Cheap	Expensive compared to a Hub

COMPARISON BETWEEN SWITCH AND ROUTER:

HARDWARE DEVICES → PARAMETERS	**SWITCH**	**ROUTER**
Layer in OSI model	Data Link Layer (L2)	Network Layer (L3)
Table	MAC Address Table	Routing Table
Address	Uses hardware MAC Addresses	Uses logical IP Addresses
Function	Connects multiple local end-devices within a single LAN segment.	Interconnects completely different networks/subnets (LAN to WAN).
No. of ports available	High density (e.g., 24 or 48 ports)	Low density (typically 2 to 4 Gigabit/FastEthernet ports)
Collisions	Eliminates collisions on individual ports via Full-Duplex.	Completely isolates collision domains across connected segments.
Transmission Mode	Full-Duplex	Full-Duplex
Broadcast Domain	Single broadcast domain per configured VLAN.	Breaks up broadcast domains (Broadcasts do not pass a router).
Speed	Very High speed (Fast switching in hardware via ASICs)	Slower than switches (Requires CPU processing for route lookups)

CONCLUSION:-

Using Cisco Packet Tracer, we analyzed how hardware operates across network layers. We observed Layer 1 Hubs and Repeaters blindly broadcast traffic, Layer 2 Switches and Bridges isolate collision domains using MAC tables, and Layer 3 Routers segment broadcast domains to connect distinct IP subnets.